10 The table shows pairs of values of x and y

x	5	6
y	400	576

(i) Tick the correct statement below.

$400 \div 5^2 = 16$ and $576 \div 6^2 = 16$.

$y \propto x$ .....

$y \propto x^2$ ..... ✓

$y \propto x^3$ .....

1 mark
Tick y proportional to x^2

(ii) Write a formula for y in terms of x

Use $y = kx^2$.

1 mark
Model $y = kx^2$

$400 = 25k$

$\div 25 \quad \div 25$

$16 = k$

1 mark
$k = 16$

Therefore $y = 16x^2$.

1 mark
$y = 16x^2$

(Total for Question 10 is 4 marks)